

The Multiplicity of Meaning

Appendix: Formal Theories of Expressing

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1 Background: higher-order logic

Given the nature of our topic, the ability to formalise our claims and arguments is a crucial check on our theorising. When talking about semantic relations in English there are constant temptations to push against the limits of what can be said literally; adding one pinch of Fregean salt after another, we end up in a pickle, with nothing to go on but loose and metaphorical formulations whose logical relationships are opaque, or set-theoretic surrogates that don't

My favoured system for formalisation is that of higher-order logic, which extends the familiar first-order logic by giving a way of forming complex predicates, and providing quantifiers that stand to expressions of arbitrary grammatical categories as the familiar first order quantifiers stand to the grammatical category of names. In this Appendix, I will use a “functional” dialect (see Dorr, Hawthorne, and Yli-Vakkuri, unpublished: ch. 0 for a gentler introduction to an alternative “relational” dialect, and Dorr 2016: §A1 for a translation-manual between the two). Every expression, or “term”, is assigned a “type” capturing its grammatical category. Types (like terms) are strings of symbols: the letters ‘ e ’ and ‘ t ’ are types, and whenever σ and τ are types and τ is not e , $(\sigma \rightarrow \tau)$ —i.e. the string comprising a left parenthesis, σ , an arrow, τ , and a right parenthesis—is another type.¹ Our *terms* are

¹There is a different system that allows for terms to have types ending in e . It raises extra questions of intelligibility—I of course think it *can* be made meaningful in many ways, but I think that there are several ways of doing so and each of them will lead to a system of logic for the broader language that's much less simple and unified than the our system below for the narrower language. That's why I'm sticking to the narrower one where e

also strings of symbols. Some special ones are designated as *variables* and *constants*, each once-and-for-all assigned a certain type; there are infinitely many variables for each type. A variable or constant of a type is a term of that type. When A is a term of some type $\sigma \rightarrow \tau$ and B is a term of type σ , (AB) is a term of type τ . And when A is a term of some type τ and v is a variable of type σ , $(\lambda v.A)$ is a term of type $\sigma \rightarrow \tau$.² Those are the only types. Terms of type τ are called *formulae*, or *sentences* when they lack free variables.

Our stock of constants will include: $\neg$ of type $t \rightarrow t$; $\wedge$ of type $t \rightarrow t \rightarrow t$; and for each type σ , $\forall_\sigma$ of type $(\sigma \rightarrow t) \rightarrow t$ and $=_\sigma$ of type $\sigma \rightarrow \sigma \rightarrow t$. When P and Q are formulae, $P \wedge Q$ abbreviates $((\wedge P)Q)$. When P is a formula and v is a variable of type σ , $\forall v P$ abbreviates $(\forall_\sigma(\lambda v.P))$. Other familiar logical constants may be added, or treated as metalinguistic abbreviations in the obvious ways. Nonlogical constants will be added depending on the topic we are theorising about: later I will introduce some constants suited to theorising about syntax and meaning.

I think I *understand* this language—or at least the fragment of the language containing only the aforementioned logical constants; obviously each particular added nonlogical constant will raise a new question of intelligibility. I'll use words like 'property' and 'proposition' in giving English glosses on its sentences, but I think it's an open question how faithful this is to the functioning of such words in ordinary English.

All the theories to be considered below will extend a certain basic *logic*, H. Its axioms and rules are given in Figure 1. This involves some controversial commitments which aren't really crucial: the theories could also be adapted in weaker systems which drastically restrict the β axiom (see Dorr, Hawthorne, and Yli-Vakkuri, unpublished: ch. 0), and perhaps also drop η . But it'll be easier to leave those weakenings to explore later.

The kind of higher-order language I have just described is convenient to actually use. But to theorise *about*, it will also be useful to have at hand a syntactically simpler but (arguably) equally expressive dialect of higher-

isn't a "terminal" type. But everything below could be straightforwardly extended to the broader system by those who, unlike me, accept the straightforward extension of the logic to that system.

²We could also, at the cost of some complexity, try to do the whole theory in a " λI " language where $(\lambda v.A)$ is well-formed only when v has at least one free occurrence in A : see Dorr 2016. This makes for some more fiddliness in the formally syntactic theory but the general situation is similar.

PC: $\vdash P$ whenever P is a tautology (substitution instance of a theorem of classical propositional logic).
UI: $\vdash \forall_\sigma F \rightarrow Fa$ (where a is of some type σ and F is a term of type $\sigma \rightarrow t$).
Ref: $\vdash A =_\sigma A$
LL: $\vdash (a = b) \rightarrow (Fa \rightarrow Fb)$
 β : $\vdash P \leftrightarrow Q$ whenever Q can be derived from P by replacing some term of the form $((\lambda v.A)b)$ with $A[v \mapsto b]$, the result of replacing every free occurrence of v in A with an occurrence of b , so long as this can be done without any free variable in b becoming bound.
 η : $\vdash P \leftrightarrow Q$ whenever Q can be derived from P by replacing some term of the form $(\lambda v.(Av))$ with A , where v is not free in A .
MP: If $\vdash P$ and $\vdash P \rightarrow Q$, then $\vdash Q$.
Gen: If $\vdash P \rightarrow Q$, and v does not occur free in P , $\vdash P \rightarrow (\forall v.Q)$.

Figure 1: Axioms and rules of H

order language, which I'll call "combinatorese". The terms of combinatorese are defined in the same way as before except that we get rid of lambda-abstraction as a way of forming complex terms. Instead, we add three extra families of logical constants. For any types ρ, σ, τ (where $\tau \neq e$) we have

$$\begin{aligned}
 S_{\rho, \sigma, \tau} &: (\rho \rightarrow \sigma \rightarrow \tau) \rightarrow (\rho \rightarrow \sigma) \rightarrow \rho \rightarrow \tau \\
 K_{\sigma, \tau} &: \tau \rightarrow \sigma \rightarrow \tau \\
 I_\tau &: \tau \rightarrow \tau^3
 \end{aligned}$$

See Hindley and Seldin 2008 for an accessible introduction and a description of the translation systems. The translation for combinatorese terms to lambda-terms is straightforward: we simply go through and everywhere make the replacements

$$\begin{aligned}
 S_{\rho \rightarrow \sigma \rightarrow \tau} &\mapsto (\lambda x. \lambda y. \lambda z. xz(yz)) \\
 K_{\sigma \rightarrow \tau} &\mapsto (\lambda y. \lambda z. y) \\
 I_\tau &\mapsto (\lambda z. z)
 \end{aligned}$$

where x, y, z are arbitrarily chosen variables of types ρ, σ, τ . The reverse

³If we want to confine ourselves to the "λI" calculus for reasons discussed in Dorr 2016, we could instead use the BCIW basis of combinators: see [*CITE].

translation, call it $*$, is defined compositionally as follows:

$$\begin{aligned}
v^* &= v && \text{where } v \text{ is a variable} \\
c^* &= c && \text{where } c \text{ is a constant} \\
(AB)^* &= (A^*B^*) \\
(\lambda v^\sigma . A^\tau)^* &= K_{\sigma, \tau} A && \text{when } v \text{ isn't free in } A \\
(\lambda v^\rho . A^\sigma B^\tau)^* &= (S_{\rho, \sigma, \tau} (\lambda v . A^*)^*) (\lambda v . B^*)^* \\
(\lambda v^\tau . v)^* &= I_\tau
\end{aligned}$$

The terms output by this translation scheme are often monstrously long and almost impossible to understand by reading them. But it can be shown (Hindley and Seldin 2008: p. ???) that when we translate into combinatoremese and back, the resulting formula is β -equivalent to the original (in the sense that the biconditional can be proved from the β -axiom and the transitivity of $\leftrightarrow$).

We have allowed variables to occur in combinatoremese terms, but they don't really do any distinctive work since we don't have variable binding. Speakers of a combinatoremese language wouldn't have any particular use for open formulae (unless they decided on one, e.g. using the variables as demonstratives). So when we are imagining giving a theory of meaning about such a community we can confine our attention to the variable-free fragment of combinatoremese.

2 Co-expressing

In Lecture 1 I suggested that semantics (in the sense of “theory of meaning”, not “model theory”) should take as its topic a certain family of *co-expressing* relations. There is one such relation for every n -tuple of types $\sigma_1, \dots, \sigma_n$; it holds between n linguistic items, n entities of those types, and a language (or linguistic community). To theorise about languages with indexicals we may want to add some further arguments (e.g. for a speaker, time, and world). But in this appendix I will ignore these for simplicity. I will also suppress the argument for a language or linguistic community: really it's ****.

Theorizing about a language that contains variables and lambda-abstraction raises certain special complexities, especially if we don't want our theory to be committed to certain controversial metaphysical claims in pure higher-orderese, such as the Functionality principle that when X and Y are entities of type type $\sigma \rightarrow \tau$ such that $Xz = Yz$ for every z of type σ , $X = Y$ (See

Dorr 2016, Bacon and Dorr, n.d. I'll explore two ways of dealing with these complexities in sections and below. But for now it will be easier to illustrate a theory of co-expressing for a *combinatores* language of the sort described in the previous section. The choice not to theorise *in* such a language as well is purely a matter of convenience, since given β there is no problem about translating everything we want to say into such a language.

3 Theory of syntax

Before we can give a formal theory of meaning for such a language, we will have to give a formal theory about its syntax. Developing such a theory involves, *inter alia*, deciding on an account of the ontology of linguistic expressions (i.e. expression-types). There are many options, the differences between which don't matter for the purposes of semantics. For example, we could develop syntax within a theory of *geometry*, identifying linguistic expressions with certain *shape-properties* (that could be instantiated by spatial regions or physical objects made of ink): for example the logical ' $\wedge$ ' would be the property of being a region shaped in that particular two-sides-of-a-triangle way. Or we could think in terms of sound-types rather than inscription-types. Or we could think of words and larger expressions as "social artifacts" and present a theory of them in terms of some more general ontological theory that also applies to other such artifacts (novels, songs, laws, fictional characters. . .). Or we could keep our lives simple by following the standard logician's practice of identifying expressions with their Gödel numbers, where natural numbers could in turn be identified with entities of some appropriate type such as $(t \rightarrow t) \rightarrow t \rightarrow t$. Since any particular such approach will involve resolving all sorts of questions of detail which don't matter when our primary interest is in semantics, it will be clearest to give the theory in a schematic form that could be filled in in any of the above ways.

So, we will suppose we have two distinguished types: χ , the type of ordinary linguistic expressions (terms), and ξ , the type of "type-labels", which are certain special linguistic items like ' t ' and ' $e \rightarrow t$ ' which play the usual role of keeping track of grammatical categories. We are neutral as regards whether χ and ξ are the same type, and which types they are: for example on the 'shape-property' approach they would both be $e \rightarrow t$. We also suppose that we have somehow or other defined the resources to characterise the way in which two terms can be combined by "application" (@) into a third term; in which two type-labels can be combined by "writing an arrow in between

them" ($\Rightarrow$) into a third type-label, and to say which terms are constants of our language having a given type-label, and which type-labels correspond to the basic types e and t . We thus have five expressions which we can treat as primitive:

$$\begin{aligned}\Rightarrow &: \xi \rightarrow \xi \rightarrow \xi \\ @ &: \chi \rightarrow \chi \rightarrow \chi \\ \text{IsConstant} &: \chi \rightarrow \xi \rightarrow t \\ \langle e \rangle &: \xi \\ \langle t \rangle &: \xi\end{aligned}$$

And we assume that we have a theory these obey the following basic axioms (omitting initial universal quantifiers):

$$\text{INJECTIVITY FOR ARROW} \quad s \Rightarrow t = s' \Rightarrow t' \rightarrow s = s' \wedge t = t'$$

$$\text{BASIC TYPES} \quad \langle e \rangle \neq s \Rightarrow t \text{ and } \langle t \rangle \neq s \Rightarrow t$$

$$\text{INJECTIVITY FOR APPLICATION} \quad a @ b = c @ d \rightarrow a = b \wedge c = d$$

$$\text{CONSTANTS} \quad \text{IsConstant}(c, s) \rightarrow c \neq a @ b$$

We can then define *being a type* as *being reachable from $\langle e \rangle$ and $\langle t \rangle$ by successive applications of $\Rightarrow$* . In higher-orderese this is expressible as follows, using universal quantification into type $\xi \rightarrow t$:

$$\text{IsType}(t) := \forall X(X\langle e \rangle \wedge X\langle t \rangle \wedge \forall s \forall s'(Xs \wedge Xs' \rightarrow X(s \Rightarrow s')) \rightarrow Xt)$$

We can't likewise define *being a term* as *being reachable from constants by successive applications of $@$* , but this would give us all sorts of ill-formed terms. Instead, we give a similar "inductive" definition of *a is a term of type s* :

$$\begin{aligned}\text{HasType}(a, t) &:= \forall X^{\xi \rightarrow \chi \rightarrow t}((\forall b \forall s(\text{IsConstant}(b, s) \rightarrow Xbs) \\ &\quad \wedge \forall b b' s s'(Xb's \wedge Xb(s \Rightarrow s')) \rightarrow X(a @ b)s') \rightarrow Xat)\end{aligned}$$

Given these definitions and the axioms the following are established by straightforward (object-language) inductive proofs:

Proposition 1. $\text{HasType}(a, s) \rightarrow \text{IsType}(s)$.

Proposition 2 (Unique typing). $\text{HasType}(a, s) \wedge \text{HasType}(a, s') \rightarrow s = s'$.

Proposition 3 (Application).

$$\text{HasType}(b, s) \rightarrow (\text{HasType}(a, s \Rightarrow t) \leftrightarrow \text{HasType}(a @ b, t))$$

We can straightforwardly extend the $\langle \cdot \rangle$ notation from the basic types e and t to all other types, by defining $\langle \sigma \rightarrow \tau \rangle = \langle \sigma \rangle \Rightarrow \langle \tau \rangle$ as a metalinguistic shorthand.

If we want to talk about the meaning of specific terms as well as the general rules for how the meanings of complex terms depend on those of their parts, we will of course need to be able to pick out certain constants. And we particularly want to be able to theorise about the very language in which we are stating our semantic and syntactic theory—or to be more exact, since we are using a lambda-language, about the combinatory language with the same constants and the special $S_{\alpha, \beta, \gamma}$ and $K_{\alpha, \beta}$ constants into which we could translate that language. For this especially interesting case, we will suppose that we have somehow or other specified a term $\langle c \rangle$ of type χ for every logical or non-logical constant of our language $\mathcal{L}$. Our basic syntactic theory C^- comprises the above four axioms together with

$$\text{IsConstant}(\langle c \rangle, \langle \sigma \rangle)$$

for each (nonlogical or logical) constant c of type σ in $\mathcal{L}$. Note that we do not assume that $\langle c \rangle$ is a *constant* for each constant c . For example, on the geometric approach, each $\langle c \rangle$ would be a complex characterisation of the shape of c , while on the Gödel-numbering approach each term $\langle c \rangle$ would be a pure combinatory term corresponding to a numeral.

We can extend the $\langle \cdot \rangle$ shorthand to other terms (well-formed and ill-formed) in the obvious way, by setting $\langle AB \rangle = \langle A \rangle @ \langle B \rangle$. Then for each well-formed combinatory term $\langle A \rangle$ of type σ , $\text{HasType}(\langle A \rangle, \langle \sigma \rangle)$ will be a theorem. Note that the identity of the term $\langle \langle A \rangle \rangle$ (of type χ) will then depend on the structure of the term $\langle A \rangle$.

We can further extend the $\langle \cdot \rangle$ shorthand to allow closed *lambda*-terms to occur inside $\langle \cdot \rangle$ just by picking a standard translation from the lambda-language into combinatores and applying that before applying $\langle \cdot \rangle$.

Finally, it will be convenient to let $\langle \cdot \rangle$ work like Quinean corner-quotes by allowing *free variables of type χ* to occur inside of $\cdot$, with the convention that $\langle a \rangle = a$ when a is such a variable. For example, $\langle \neg a \rangle$ abbreviates $\langle \neg \rangle @ \langle a \rangle$ which abbreviates $\langle \neg \rangle @ a$. Note that the thing inside the angle-brackets, $\neg a$,

is not even a typable term (since $\neg$ is a constant of type $t \rightarrow t$ while a is a variable of type χ which is presumably distinct from t). That's not a problem; the general notion of "term" that allows ill-typed terms makes perfect sense too.

4 Semantic theory

We have now described a family of higher-order languages capable of talking about the simple syntax of combinatory higher-order languages. Our semantic theory will be given in an unspecified language $\mathcal{L}$ which contains in addition to the standard logical constants and some unspecified nonlogical constants (in terms of which the syntactic vocabulary above is somehow define), an additional family of "co-expression" constants $E_{\sigma_1, \dots, \sigma_n}$ of type $\chi^n \rightarrow \sigma_1 \rightarrow \dots \rightarrow \sigma_n \rightarrow t$. (For ease of exposition, I am suppressing the extra argument for a language / linguistic community, as well as the further arguments that would be useful if we wanted to give a more discriminating treatment of indexicals.) Our basic meaning-theory $\mathbf{C}$ will add the following six axiom-schemes concerning E to the syntactic theory described in the previous section. Here $\vec{a} = a_1, \dots, a_n$ are any variables of type χ , $\vec{\sigma} = \sigma_1, \dots, \sigma_n$ are any types, and $\vec{x} = x_1, \dots, x_n$ are any variables of types $\sigma_1, \dots, \sigma_n$ respectively.

PERMUTATION $\vec{a} E_{\vec{\sigma}} \vec{x} \rightarrow a_{k_1}, \dots, a_{k_m} E_{\sigma_{k_1}, \dots, \sigma_{k_m}} x_{k_1}, \dots, x_{k_m}$ for any $k_1, \dots, k_m \leq n$

REPETITION $a, a E_{\sigma, \sigma} x, y \rightarrow x = y$

WELL-FORMEDNESS $a E_{\sigma} x \rightarrow \text{HasType}(a, \langle \sigma \rangle)$

INTERPRETABILITY $\text{HasType}(b, \langle \tau \rangle) \wedge \vec{a} E_{\vec{\sigma}} \vec{x} \rightarrow \exists y (\vec{a}, b E_{\vec{\sigma}, \tau} \vec{x}, y)$

COMPOSITIONALITY $a, b, a @ b E_{\sigma \rightarrow \tau, \sigma, \tau} x, y, z \rightarrow z = xy$

LOGICAL CONSTANTS $\langle c \rangle E_{\sigma} x \rightarrow x = c$ for each *logical* constant c of type σ .

Note that all the $\rightarrow$ s here can provably be upgraded to $\leftrightarrow$ s, except for those instances of Permutation where not all of the numbers $1, \dots, n$ are among $k_1, \dots, k_m$. What counts as a "logical" constant doesn't really matter so long as the $E_{\vec{\sigma}}$ constants don't count; to make the theory as strong as possible we will assume that all the constants that occur in the syntactic primitives from the last section count as logical.

The status of the axioms is rather different. Permutation and Repetition are plausibly necessary truths which would apply for any value of the

suppressed “linguistic community” argument-place. Well-formedness, Interpretability and Compositionality are plausibly contingent, and would be false if our community didn’t use the operation described by @ as their way of forming complex expressions; Interpretability would also be false if they didn’t use the same expressions as constants, or used those expressions as constants but with a different convention about their types. And Logical Constants is also contingent, and its truth depends not only on those relatively general facts about the community, but on its using the specific expressions that are logical constants in the way that makes them express what they do: there’s a whole literature in metasemantics devoted specifically to questions like ‘How does a community need to treat a symbol in order for that symbol to express *conjunction* in its language?’

We will later consider some strengthenings of $\mathbf{C}$, but it is worth first exploring the theory as it stands.

5 Some theorems of $\mathbf{C}$

Proposition 4 (Compositionality).

$$\text{HasType}(b, \langle \sigma \rangle) \rightarrow a @ b, \vec{c} \mathbf{E}_{\tau, \vec{\sigma}} x, \vec{y} \leftrightarrow \exists z \exists z' (x = zz' \wedge a, b, \vec{c} \mathbf{E}_{\sigma, \vec{\sigma} \rightarrow \tau, \vec{\sigma}} z, z', \vec{y})$$

Proposition 5 (Repetition).

$$\vec{a}, \vec{a}, b \mathbf{E} \vec{x}, \vec{y}, z \leftrightarrow \vec{a}, b \mathbf{E} \vec{x}, z \wedge \bigwedge_{i \leq n} x_i = y_i$$

Proposition 6 (Thinning). For any types $\sigma_1, \dots, \sigma_n$ and any $k_1, \dots, k_m \leq n$:

$$\begin{aligned} & \left(\bigwedge_{i \leq n} \text{HasType}(a_i, \langle \sigma_i \rangle) \wedge a_{k_1}, \dots, a_{k_m} \mathbf{E}_{\sigma_{k_1}, \dots, \sigma_{k_m}} x_1, \dots, x_m \right) \\ & \leftrightarrow \exists y_1 \dots y_n (a_1, \dots, a_n \mathbf{E}_{\sigma_1, \dots, \sigma_n} y_1, \dots, y_n \wedge \bigwedge_{i \leq m} x_i = y_{k_i}) \end{aligned}$$

Proof. Left to right: [***FILL IN].

Right to left: using the right-to-left direction of Interpretability and Permutation. $\square$

Proposition 7 (Constants). For any logical constants $\vec{c}$,

$$\vec{a}, \langle \vec{c} \rangle \in \vec{x}, \vec{y} \leftrightarrow \vec{a} \in \vec{x} \wedge \bigwedge_i y_i = c_i$$

Proposition 8 (Analysis). For any open or closed combinatory term A of type σ with no non-logical constants and free variables among $\vec{z} = z_1, \dots, z_n$ of types $\vec{\sigma} = \sigma_1, \dots, \sigma_n$; any variables $\vec{b} = b_1, \dots, b_n$ and $\vec{c} = c_1, \dots, c_m$ of type χ , and any types $\vec{\tau} = \tau_1, \dots, \tau_m$:

$$\begin{aligned} (*) \quad & \left(\bigwedge_{i \leq n} \text{HasType}(b_i, \langle \sigma_i \rangle) \wedge \langle A[b_i/z_i] \rangle, \vec{c} \in_{\sigma, \vec{\tau}} x, \vec{y} \right) \\ & \leftrightarrow \exists \vec{z} (\vec{b}, \vec{c} \in_{\vec{\sigma}, \vec{\tau}} \vec{z}, \vec{y} \wedge x = A) \end{aligned}$$

Proof. By induction on the complexity of A , for fixed $n, m, \vec{z}, \vec{b}, \vec{c}$.

Base case (i): A is a variable z_k . Then the instance of (*) is

$$\left(\bigwedge_{i \leq n} \text{HasType}(b_i, \langle \sigma_i \rangle) \wedge b_k, \vec{c} \in_{\sigma, \vec{\tau}} x, \vec{y} \right) \leftrightarrow \exists \vec{z} (\vec{b}, \vec{c} \in_{\vec{\sigma}, \vec{\tau}} \vec{z}, \vec{y} \wedge x = z_k)$$

which follows from Proposition 6.

Base case (ii): A is a logical constant k ; then the instance of (*) is

$$\left(\bigwedge_{i \leq n} \text{HasType}(b_i, \langle \sigma_i \rangle) \wedge \langle k \rangle, \vec{c} \in_{\sigma, \vec{\tau}} x, \vec{y} \right) \leftrightarrow \exists \vec{z} (\vec{b}, \vec{c} \in_{\vec{\sigma}, \vec{\tau}} \vec{z}, \vec{y} \wedge x = k)$$

which follows from Proposition 7 and Interpretability.

Inductive step. Suppose A is BC where $B : \tau \rightarrow \sigma$ and $C : \tau$ satisfy (*). Then

$A[b_i/z_i]$ is $B[b_i/z_i]C[b_i/z_i]$, and so $\langle A[b_i/z_i] \rangle$ is $\langle B[b_i/z_i] \rangle @ \langle C[b_i/z_i] \rangle$. So,

$$\begin{aligned}
& \langle A[b_i/z_i] \rangle, \vec{c} E_{\sigma, \vec{\tau}} x, \vec{y} \\
& \leftrightarrow \langle B[b_i/z_i] \rangle @ \langle C[b_i/z_i] \rangle, \vec{c} E_{\sigma, \vec{\tau}} x, \vec{y} \\
& \leftrightarrow \exists z \exists z' (\langle B[b_i/z_i] \rangle, \langle C[b_i/z_i] \rangle, \vec{c} E_{\tau \rightarrow \sigma, \tau, \vec{\tau}} z, z', \vec{y} \wedge x = zz') & \text{Composition} \\
& \leftrightarrow \exists z \exists z' \exists \vec{z} (\vec{b} \langle C[b_i/z_i] \rangle \vec{c} E_{\vec{\sigma}, \tau, \vec{\tau}} \vec{z}, z', \vec{y} \wedge x = zz' \wedge z = B) & \text{IH for } B \\
& \leftrightarrow \exists z' \exists \vec{z} (\vec{b} \langle C[b_i/z_i] \rangle, \vec{c} E_{\vec{\sigma}, \tau, \vec{\tau}} \vec{z}, z', \vec{y} \wedge x = Bz') & \text{logic} \\
& \leftrightarrow \exists z' \exists \vec{z} (\langle C[b_i/z_i] \rangle, \vec{b}, \vec{c} E_{\tau, \vec{\sigma}, \vec{\tau}} z', \vec{z}, \vec{y} \wedge x = Bz') & \text{Perm} \\
& \leftrightarrow \exists z' \exists \vec{z} \exists \vec{u} (\vec{b}, \vec{b}, \vec{c} E_{\vec{\sigma}, \vec{\sigma}, \vec{\tau}} \vec{u}, \vec{z}, \vec{y} \wedge x = Bz' \wedge z' = C[u_i/z_i]) & \text{IH for } C \\
& \leftrightarrow \exists \vec{z} \exists \vec{u} (\vec{b}, \vec{b}, \vec{c} E_{\vec{\sigma}, \vec{\sigma}, \vec{\tau}} \vec{u}, \vec{z}, \vec{y} \wedge x = BC[u_i/z_i]) & \text{logic} \\
& \leftrightarrow \exists \vec{z} (\vec{b}, \vec{c} E_{\vec{\sigma}, \vec{\tau}} \vec{z}, \vec{y} \wedge x = BC) & \text{lemma (Repetition)}
\end{aligned}$$

□

Proposition 9. When A is a closed term of type σ with only logical constants, $\langle A \rangle E_{\sigma} x \leftrightarrow x = A$

Proof. This is the special case of Proposition 8 where $n = m = 0$. □

Proposition 10. If P is an open or closed theorem containing no nonlogical constants and free variables among $\vec{z}$ of types $\vec{\sigma}$, and $\vec{b}$ are variables of type χ , then

$$\forall p (\langle P[b_i/z_i] \rangle E_t p \rightarrow p)$$

Proof. By Proposition 8,

$$\langle P[b_i/z_i] \rangle E_t p \rightarrow \exists \vec{z} (\vec{b} E_{\vec{\sigma}} \vec{z} \wedge p = P).$$

But if P is a theorem, $(p = P) \rightarrow p$ is one too. □

Proposition 11. If $P_1 \rightarrow \dots \rightarrow P_n \rightarrow P_{n+1}$ is an open or closed theorem containing no nonlogical constants and free variables among $\vec{z}$ of types $\vec{\sigma}$, and $\vec{b}$ are variables of type χ , then

$$\begin{aligned}
& \forall p_1 \dots p_n q (\langle P_1[b_i/z_i] \rangle, \dots, \langle P_{n+1}[b_i/z_i] \rangle, E_{t^{n+1}} p_1, \dots, p_{n+1} \rightarrow \\
& \quad p_1 \rightarrow \dots \rightarrow p_n \rightarrow p_{n+1})
\end{aligned}$$

Proof. By Proposition 8,

$$\begin{aligned} \langle P_1[b_i/z_i] \rangle, \dots, \langle P_{n+1}[b_i/z_i] \rangle, E_{t^{n+1}} p_1, \dots, p_{n+1} \\ \rightarrow \langle (P_1 \rightarrow \dots \rightarrow P_{n+1})[b_i/z_i] \rangle E_t (p_1 \rightarrow \dots \rightarrow p_{n+1}) \end{aligned}$$

and by the previous lemma,

$$\langle (P_1 \rightarrow \dots \rightarrow P_{n+1})[b_i/z_i] \rangle E_t (p_1 \rightarrow \dots \rightarrow p_{n+1}) \rightarrow (p_1 \rightarrow \dots \rightarrow p_{n+1}) \quad \square$$

6 The truth theory in $\mathbf{C}$

To see how $\mathbf{C}$ relates to the classical truth theories that have been explored in the literature on truth, let us define:

$$\begin{aligned} \text{True}(a) &:= \exists p(a \ E_t \ p) \wedge \forall p((a \ E_t \ p) \rightarrow p) \\ \text{TrueOf}(a, x_1^{\sigma_1}, \dots, x_n^{\sigma_n}) &:= \exists Y(a \ E_{\sigma_1 \rightarrow \dots \rightarrow \sigma_n \rightarrow t} \ Y) \wedge \\ &\quad \forall Y((a \ E_{\sigma_1 \rightarrow \dots \rightarrow \sigma_n \rightarrow t} \ Y) \rightarrow Yx_1 \dots x_n) \end{aligned}$$

This is analogous to the definition (TDEF3) of utterance-truth mentioned in Lecture 1.

Proposition 12. Whenever P is a closed theorem of $\mathbf{C}^-$ (our base theory of syntax), $\vdash_{\mathbf{C}} \text{True}\langle P \rangle$.

Proof. First of all, when P is a sentence, $\text{HasType}(\langle P \rangle, \langle t \rangle)$ is a theorem, and thus by Interpretability so is $\exists q(\langle P \rangle \ E_t \ q)$. So it suffices to show that $\forall q((\langle P \rangle \ E_t \ q) \rightarrow q)$. To do so, note that since all the distinctive axioms of $\mathbf{C}^-$ involve only “logical” vocabulary, whenever P is a theorem of $\mathbf{C}^-$, there is some (open or closed) theorem Q containing only “logical” vocabulary such that P is $Q[A_i/z_i]$ for some closed terms $A_1, \dots, A_n$ of certain types $\sigma_1, \dots, \sigma_n$ and variables $z_1, \dots, z_n$ of those types. Since $\text{HasType}(\langle A_i \rangle, \langle \sigma_i \rangle)$ is a theorem for each i , Proposition 8 yields:

$$\langle Q[\langle A_i \rangle/z_i] \rangle E_t p \rightarrow \exists \vec{z}(\langle A_1 \rangle, \dots, \langle A_n \rangle E_{\vec{\sigma}} \vec{z} \wedge p = P)$$

But since the formula Q is a theorem of $\mathbf{C}^-$, $(p = Q) \rightarrow p$ is too, and so we have

$$\langle Q[\langle A_i \rangle/z_i] \rangle E_t p \rightarrow p$$

□

Proposition 13 (Closure). $\text{True}\langle a \rightarrow b \rangle \rightarrow (\text{True}(a) \rightarrow \text{True}(b))$.

Proof. Suppose (i) $\text{True}\langle a \rightarrow b \rangle$, (ii) $\text{True}(a)$, and (iii) $b \text{ E}_t p$. By (iii) and Interpretability we have

$$\exists q \exists r (\langle a \rightarrow b \rangle, a, b \text{ E}_{t,t,t} r, q, p)$$

By Proposition 11, we have

$$\forall q \forall r ((\langle a \rightarrow b \rangle, a, b \text{ E}_{t,t,t} r, q, p) \rightarrow r \rightarrow q \rightarrow p)$$

By Reordering,

$$\forall q \forall r ((\langle a \rightarrow b \rangle, a, b \text{ E}_{t,t,t} r, q, p) \rightarrow \langle a \rightarrow b \rangle \text{ E}_t r \wedge a \text{ E}_t q)$$

and by assumptions (i) and (ii),

$$\forall q \forall r ((\langle a \rightarrow b \rangle \text{ E}_t r) \wedge (a \text{ E}_t q) \rightarrow (r \wedge q)) \quad \square$$

Proposition 14 (T-Cons). $\text{True}\langle \neg a \rangle \rightarrow \neg \text{True}(a)$.

Proof. Suppose $\text{True}\langle \neg a \rangle$ and $a \text{ E}_t p$. Then by Proposition 7, $a, \langle \neg \rangle \text{ E}_{t,t \rightarrow t} p, (\neg)$. So by Compositionality, $\langle \neg a \rangle \text{ E}_t \neg p$. So $\neg p$ by the assumption that $\text{True}\langle \neg a \rangle$. So, we have $\forall p (a \text{ E}_t p \rightarrow \neg p)$. But since $\exists p (\langle \neg a \rangle \text{ E}_t p)$, we have $\text{HasType}(\langle \neg a \rangle, t)$ by Well-formedness, and hence $\text{HasType}(a, t)$ by Proposition 3, and so $\exists p (a \text{ E}_t p)$ by Interpretability. So $\exists p ((a \text{ E}_t p) \wedge \neg p)$, and hence $\neg \text{True}(a)$. $\square$

Proposition 15 (U-inf). $\forall x^\sigma (\text{TrueOf}(a, x)) \rightarrow \text{True}\langle \forall_\sigma a \rangle$.

Proof. Suppose $\forall x^\sigma (\text{TrueOf}(a, x))$. Then $\exists x^\sigma (\text{TrueOf}(a, x))$, so $\exists Y (a \text{ E}_{\sigma \rightarrow t} Y)$, and thus $\exists p (\langle \forall_\sigma a \rangle \text{ E}_t p)$ by Interpretability and Compositionality. By Proposition 8,

$$\forall p (\langle \forall_\sigma a \rangle \text{ E}_t p \rightarrow \exists Y (a \text{ E}_{\sigma \rightarrow t} Y \wedge p = \forall_\sigma Y))$$

But by the assumption that $\forall x^\sigma (\text{TrueOf}(a, x))$, we have $\forall Y (a \text{ E}_{\sigma \rightarrow t} Y \rightarrow \forall_\sigma Y)$. Thus

$$\forall p (\langle \forall_\sigma a \rangle \text{ E}_t p \rightarrow \exists Y (\forall_\sigma Y \wedge p = \forall_\sigma Y))$$

and thus $\forall p (\langle \forall_\sigma a \rangle \text{ E}_t p \rightarrow p)$. $\square$

Thus under this definition of True, $\mathbf{C}$ includes the “Base” theory from Friedman and Sheard 1987—namely, $\text{True}(P)$ for every theorem of the underlying syntactic theory (for which they use Peano arithmetic), plus closure of truth under modus ponens. It also includes (analogues of) their principles T-Cons and U-Inf.

7 Adding the disquotational expressing schema

Let $\mathbf{C}^+$ be the result of adding every instance of the following schema to $\mathbf{C}$, where $A_1, \dots, A_n$ are any closed terms of types $\sigma_1, \dots, \sigma_n$:

$$(DE) \quad \langle A_1 \rangle \dots \langle A_n \rangle E_{\sigma_1, \dots, \sigma_n} A_1 \dots A_n$$

It is easy to show that given $\mathbf{C}$, all instances of DE follow from the instances where $A_1 \dots A_n$ are nonlogical constants.

With True defined as before, $\mathbf{C}^+$ proves every instance of the principle

$$\mathbf{T-Out} \quad \text{True}\langle P \rangle \rightarrow P$$

where P is a closed sentence. For by the definition of True we have $\text{True}(\langle P \rangle) \rightarrow ((\langle P \rangle E_t P) \rightarrow P)$, and by DE we have $\langle P \rangle E_t P$.⁴

Let Het be (the combinatory term that translates) $\lambda y. \exists Z (y E_{\xi \rightarrow t} Z \wedge \neg Zy)$.⁵ By DE,

$$\langle \text{Het} \rangle E_{\xi \rightarrow t} \text{Het}$$

. Given the predicational variant of Prior’s theorem (see footnote **), this

⁴Putting this together with the principles about truth from the previous section (which are already theorems of $\mathbf{C}$), we see that with this definition of True, $\mathbf{C}^+$ proves all but one of the characteristic principles of “System H” from Friedman and Sheard 1987. The one that isn’t a theorem is

$$\mathbf{T-Rep} \quad \text{True}\langle P \rangle \rightarrow \text{True}\langle \text{True}\langle P \rangle \rangle.$$

If we wanted to, we could strengthen to theory to prove every instance of this schema by adding the principle that all of the relations expressed by one of the co-expressing predicates $\langle E \rangle_{\vec{\sigma}}$ are coextensive with or extensionally narrower than expressing itself:

$$\mathbf{Shrinking} \quad \langle E_{\vec{\sigma}} \rangle E_{\chi^n, \vec{\sigma}} R \rightarrow \forall \vec{a} \vec{x} (R \vec{a} \vec{x} \rightarrow \vec{a} E_{\vec{\sigma}} \vec{x})$$

I don’t know whether this is consistent; certainly it does not hold in the models discussed below. But in any case, I don’t think it is philosophically well-motivated.

⁵Note that this would be equivalent to $\lambda y. \neg \text{TrueOf}(y, y)$ if we defined ‘True’ by (TDEF2).

logically implies

$$(1) \quad \exists Z(\langle \text{Het} \rangle E_{\xi \rightarrow t} Z \wedge \neg Z \langle \text{Het} \rangle)$$

and thus

$$(2) \quad \text{Het} \langle \text{Het} \rangle$$

This shows that some obvious strengthenings of C^+ are inconsistent. Most obviously, one gets inconsistency by adding the following:

$$\textbf{Uniqueness} \quad \forall a \forall x \forall y (a E x \wedge a E y \rightarrow x = y)$$

But clearly, $C^+ + \text{Uniqueness}$ is inconsistent, since from 1 and 2 we have

$$(3) \quad \exists Z \exists Z' (\langle \text{Het} \rangle E_{\xi \rightarrow t} Z \wedge \langle \text{Het} \rangle E_{\xi \rightarrow t} Z' \wedge Z \neq Z')$$

One could also consider adding all sentences of the form $\text{True}(P)$, where P is a theorem of C^+ . But we can't consistently do this either: in particular we can't consistently add $\text{True}(\langle \text{Het} \langle \text{Het} \rangle \rangle)$, since its negation is already a theorem of C ! That's because C proves that quote-names uniquely refer, i.e. all instances of the following schema

$$(4) \quad \vdash_C \forall x (\langle \langle A \rangle \rangle E_x x \leftrightarrow x = \langle A \rangle)$$

(by Proposition 9) and as a result we also have

$$(5) \quad \vdash_C \forall a (\text{True } a @ \langle \langle B \rangle \rangle \leftrightarrow \text{TrueOf}(a, \langle B \rangle))$$

so in particular

$$(6) \quad \text{True}(\langle \text{Het} \langle \text{Het} \rangle \rangle) \leftrightarrow \text{TrueOf}(\langle \text{Het} \rangle, \langle \text{Het} \rangle)$$

and thus

$$(7) \quad \neg \text{True}(\langle \text{Het} \langle \text{Het} \rangle \rangle) \leftrightarrow \text{Het} \langle \text{Het} \rangle$$

8 Infinitely many things expressed

We have seen that C^+ implies that some terms express more than one thing. In this section we will strengthen this by finding terms for which C^+ implies

that they express infinitely many things. Our approach is loosely inspired by Berry's paradox (of the least number not "definable" in fewer than nineteen syllables).

Let ν be a type with $\text{Num} : \nu \rightarrow t$, $\text{Suc} : \nu \rightarrow \nu$, $\leq : \nu \rightarrow \nu \rightarrow t$ obeying the second-order Peano axioms.

A term 'determines' a number iff it expresses a relation that it uniquely bears to that number:

$$D := \lambda a \lambda n. \exists R ((a \text{ E}_{\chi \rightarrow \nu \rightarrow t} R) \wedge \forall m (Ram \leftrightarrow m = n))$$

A term 'Berries' a number iff it is the least number it doesn't determine:

$$B := \lambda a \lambda n. \forall m (m \leq n \rightarrow (Dan \leftrightarrow m \neq n))$$

We can show that the term B determines every number. For if it didn't, it would Berry exactly one number. But then it would determine that number, since by DE it expresses the Berrying relation which it uniquely bears to that number: contradiction.

Formally:

1. $\forall a \forall n (Ban \rightarrow \forall m (Bam \leftrightarrow m = n))$	Reflexivity of $\leq$
2. $\forall n (Ban \rightarrow \neg Dan)$	Def. B
3. $\forall n \neg Ban \rightarrow \forall n (\text{Num } n \rightarrow Dan)$	Least number principle
4. $\forall a (a \text{ E}_{\chi \rightarrow \nu \rightarrow t} B \rightarrow \forall n (Ban \rightarrow Dan))$	(1), Def. D
5. $\forall a (a \text{ E}_{\chi \rightarrow \nu \rightarrow t} B \rightarrow \forall n \neg Ban)$	(2, 4)
6. $\forall a (a \text{ E}_{\chi \rightarrow \nu \rightarrow t} B \rightarrow \forall n (\text{Num } bn \rightarrow Dan))$	(3, 5)
7. $\langle B \rangle \text{ E}_{\chi \rightarrow \nu \rightarrow t} B$	DE
8. $\forall n (\text{Num } n \rightarrow D \langle B \rangle n)$	(6, 7)

So, B must express infinitely many non-coextensive relations, since for each of the infinitely many numbers there is a relation that it expresses and bears uniquely to that number. Since B is defined entirely in terms of logical constants and $\text{E}_{\chi \rightarrow \nu \rightarrow t}$, it follows that $\text{E}_{\chi \rightarrow \nu \rightarrow t}$ also expresses infinitely many non-coextensive relations.

9 Extensions of $\mathbf{C}$ and $\mathbf{C}^+$

We just saw that adding $\text{True}(P)$ whenever $\vdash_{\mathbf{C}^+} P$ to $\mathbf{C}^+$ would lead to inconsistency. In Lecture 3, I discussed why this shouldn't be regarded as a philosophical problem. But there are weaker extensions that are provably consistent and seem quite well motivated. For example, we could add $\text{True}(P)$ whenever $\vdash_{\mathbf{C}} P$. It seems plausible that the axioms of $\mathbf{C}$ (and hence also its theorems) exclusively express truths. Those axioms seem, after all, to be fairly basic facts about the workings of co-expression that one might plausibly think of as providing “constraints” on the relations expressed by the $\mathbf{E}_{\mathcal{E}}$ predicates, even given that DE-instances don't have the same status.

We could go further in the same direction by adding $\text{True}(\langle \text{True}(P) \rangle)$ whenever $\vdash_{\mathbf{C}} P$, and so forth. Define:

$$\begin{aligned} \mathbf{C}_0 &= \mathbf{C} \\ \mathbf{C}_{n+1} &= \mathbf{C}_n + \{\text{True}(\langle P \rangle) \mid \mathbf{C}_n \vdash P\} \end{aligned}$$

and

$$\mathbf{C}_n^+ = \mathbf{C}_n + DE.$$

We can also consider theories $\mathbf{C}_n^+$ and $\mathbf{C}_\omega^+$ which add DE to $\mathbf{C}_n$ and $\mathbf{C}_\omega$. This yields two sequences of increasingly strong theories, all consistent as we will see in the next section.

What about the limits of those sequences?

$$\begin{aligned} \mathbf{C}_\omega &= \bigcup_n \mathbf{C}_n \\ \mathbf{C}_\omega^+ &= \bigcup_n \mathbf{C}_n^+ = \mathbf{C}_\omega + DE \end{aligned}$$

$\mathbf{C}_\omega^+$ might initially seem attractive; after all it feels a bit unmotivated to stop at $\mathbf{C}_3^+$ or $\mathbf{C}_{17}^+$.

Unfortunately, it is pathological. As a corollary of by McGee's theorem (**McGeeHTPB**), not only $\mathbf{C}_\omega^+$ but the weaker $\mathbf{C}_\omega$ is ω -inconsistent: when we define ‘natural number’ and each particular numeral $\langle n \rangle$ in some standard way, we can find a term Φ such that $\neg\Phi\langle n \rangle$ is a theorem for each n , although $\exists n(\text{Num } n \wedge \Phi n)$ is also a theorem. This follows from the fact that $\mathbf{C}_\omega$ is closed under the rule ‘T-Intro’—i.e. it contains $\text{True}(\langle P \rangle)$ whenever it contains P —

and as we have already seen, it includes T-Closure, T-Cons, U-inf. McGee’s theorem establishes the ω -inconsistency of any theory with these properties.

Should we be worried by this result? I don’t think so. In fact, I don’t see any good reason to think that there is *any sentence at all* — even $\forall x(x = x)$ — such that all the sentences we can get by successively applying True to that sentence are themselves True (with True defined as above). Different interpretations of ‘express’ will correspond to different interpretations of ‘literal’. And when we start with the collection of literal utterances and successively broaden it by including first all the utterances that are not literal but have some property that could be literally expressed by ‘literal’, and next all the utterances that have no such property but still have some property that could be expressed by ‘have some property that could be literally expressed by ‘literal’’, and so on, it seems quite natural to suppose that we will eventually find ourselves with a collection of utterances so big as to include flagrantly nonliteral uses such as one where someone says ‘ $\forall x(x = x)$ ’ to mean that snow is black. For more on this, see Dorr 2010.⁶

You might think that questions like ‘How many times does one have to apply “True” to $\forall x(x = x)$ to get a sentence that isn’t True would be somehow especially embarrassing for me. But I am free to say that this question is quite vague, i.e. that ‘is the least n such that $\neg \text{True}^n(\forall x(x = x))$ ’ expresses many different properties which span a wide range of numbers. And the question where this range begins and ends can itself be vague in the same way.

10 Consistency proofs

We want to show the consistency, and ω -consistency, of C_n^+ for every n . Note that the consistency of C_ω^+ follows immediately from the consistency of each C_n^+ by the compactness theorem. However, we already know from the previous section that neither C_ω^+ nor C_ω is ω -consistent.

- (i) We fix a base model $\mathfrak{B}$ of the syntactic base theory C^- that only interprets the language $\mathcal{L}^-$ (without any of the E_σ predicates). We

⁶This is one of many structural similarities between the present family of theories and those discussed in **BaconRAD**: Bacon too has endorses some initial segment of a countable sequence of stronger and stronger theories, but does not endorse them all since their union is ω -inconsistent. In the final version of this appendix I want to include a detailed comparison of Bacon’s theories (which have to do with families of one-one “expressing” functions for each type rather than a single one-many “expressing” relation for each type) with mine.

will choose $\mathfrak{B}$ to be a “standard” — Fregean, full, and functional — model with basic types χ, ξ , and t . Let D_σ be its domain of type σ . (Note that the truth of $\mathbf{C}^-$ requires that D_χ and D_ξ are infinite, whereas being Fregean means that $D_t = \{0, 1\}$.)

We fix the interpretation of `IsConstant` in $\mathfrak{B}$ so that for $x \in D_\chi$ and $s \in D_\xi$, $\llbracket \text{IsConstant} \rrbracket_{\mathfrak{B}}(x, s) = 1$ only when there is an $\mathcal{L}$ -constant c of some type σ such that $x = \llbracket \langle c \rangle \rrbracket_{\mathfrak{B}}$ and $s = \llbracket \langle \sigma \rangle \rrbracket_{\mathfrak{B}}$. Given the standardness of the model, it follows that $\llbracket \text{HasType} \rrbracket_{\mathfrak{B}}(x, s) = 1$ iff for some $\mathcal{L}$ -term A of some type σ , x is $\llbracket \langle A \rangle \rrbracket$ and s is $\llbracket \langle \sigma \rangle \rrbracket$.

- (ii) Let B be the set of all models for $\mathcal{L}$ that extend $\mathfrak{B}$ with some arbitrary interpretations of the predicates $\mathbf{E}_{\vec{\sigma}}$.

Note that since every model in B is *standard*, any theory that is true in some model in B is ω -consistent.

- (iii) When $\mathfrak{M}, \mathfrak{M}' \in B$, say that $\mathfrak{M} \leq \mathfrak{M}'$ iff for every sequence of types $\vec{\sigma}$, the extension of $\mathbf{E}_{\vec{\sigma}}$ in $\mathfrak{M}$ is a subset of its extension in $\mathfrak{M}'$.
- (iv) For each $S \subseteq B$, we associate a single model $\mathfrak{M}_S \in B$ with the following interpretations of the co-expression predicates:

$$\llbracket \mathbf{E}_{\sigma_1 \dots \sigma_n} \rrbracket_{\mathfrak{M}_S}(x_1, \dots, x_n, y_1, \dots, y_n) = 1 \text{ (where } x_1, \dots, x_n \in D_\chi, y_1 \in D_{\sigma_1}, \dots, y_n \in D_{\sigma_n}) \text{ iff there are closed } \mathcal{L}\text{-terms } A_1, \dots, A_n \text{ of types } \sigma_1, \dots, \sigma_n \text{ and some } \mathfrak{M} \in S \text{ such that for each } i, x_i = \llbracket \langle A_i \rangle \rrbracket \text{ and } y_i = \llbracket \langle A_i \rangle \rrbracket.$$

Note that this operation is monotonic, in the sense that whenever $S \subseteq S'$, $\mathfrak{M}_S \leq \mathfrak{M}_{S'}$.

- (v) For every $\mathfrak{M} \in B$, we associate a nonempty set $T_{\mathfrak{M}} \subseteq B$, as follows:

$$T_{\mathfrak{M}} = \{\mathfrak{M}' : \mathfrak{M}' \models \mathbf{C} \text{ and } \mathfrak{M}' \leq \mathfrak{M}\}$$

Note that this operation is monotonic, in the sense that whenever $\mathfrak{M} \leq \mathfrak{M}'$, $T_{\mathfrak{M}} \subseteq T_{\mathfrak{M}'}$. This follows from the transitivity of $\leq$.

- (vi) Composing these two operations gives a “jump” operation $J : B \rightarrow B$. For each $\mathfrak{M} \in B$, $J\mathfrak{M}$ is $\mathfrak{M}_{T_{\mathfrak{M}}}$.

This operation is also monotonic since it is the composition of two monotonic operations: if $\mathfrak{M} \leq \mathfrak{M}'$, $J\mathfrak{M} \leq J\mathfrak{M}'$.

- (vii) Let $\mathfrak{M}_0$ be $\mathfrak{M}_B$.

Let $\mathfrak{M}_{n+1} = J\mathfrak{M}_n$.

Notice that since $T_{\mathfrak{M}_0} \subseteq B$, by monotonicity $\mathfrak{M}_{T_{\mathfrak{M}_0}} \leq \mathfrak{M}_B$, i.e. $\mathfrak{M}_1 = \mathfrak{M}_0$.
So by the monotonicity of J , $\mathfrak{M}_0 \geq \mathfrak{M}_1 \geq \dots$

Proposition 16. If $S \subseteq B$ is nonempty, then $\mathfrak{M}_S \models C$.

Proof. Reordering and Repetition are immediate just from the definition of $\mathfrak{M}_S$.

Well-formedness holds because if $\llbracket E_\sigma \rrbracket_{\mathfrak{M}_S}(x, y) = 1$, there is a closed term A of type σ such that $x = \llbracket \langle A \rangle \rrbracket$, but then $\llbracket \text{HasType} \rrbracket_{\mathfrak{M}_S}(x, \llbracket \langle \sigma \rangle \rrbracket) = \llbracket \text{HasType} \rrbracket_{\mathfrak{B}}(x, \llbracket \langle \sigma \rangle \rrbracket) = 1$ since $\mathfrak{B}$ models C^- and $C^- \vdash \text{HasType}(\langle A \rangle, \langle s \rangle)$.

For interpretability, note that if $\text{HasType}(x, \llbracket \langle \sigma \rangle \rrbracket) = 1$, then there is some $\mathcal{L}$ -term A .

□

Corollary 17. $\mathfrak{M}_n \models C$ for every n .

Proposition 18. If $\mathfrak{M}_S \in S$ then every instance of DE is true in $\mathfrak{M}_S$.

Proof. obvious.

□

Proposition 19. Every instance of DE is true in each $\mathfrak{M}_n$.

Proof. $\mathfrak{M}_0 \in S_0$ since S_0 contains every B -model. And $\mathfrak{M}_{n+1} = \mathfrak{M}_{S_{\mathfrak{M}_n}} \in S_{\mathfrak{M}_n}$, since $\mathfrak{M}_{n+1} \leq \mathfrak{M}_n$ and $\mathfrak{M}_{n+1} \models C$ by Corollary 17.

□

Proposition 20. If $\mathfrak{M} \models P$ for every $\mathfrak{M} \in S$, then $\mathfrak{M}_S \models \text{True}\langle P \rangle$.

Proposition 21. If $\mathfrak{M} \leq \mathfrak{M}'$, $\mathfrak{M} \models C$, and $\mathfrak{M}' \models \text{True}\langle P \rangle$ then $\mathfrak{M} \models \text{True}\langle P \rangle$.

Proof. Suppose the antecedent holds. Since $\mathfrak{M}' \models \text{True}\langle P \rangle$, $\mathfrak{M}' \models \forall p((\langle P \rangle E_t p) \rightarrow p)$, so $\llbracket E_t \rrbracket_{\mathfrak{M}'}(\llbracket \langle P \rangle \rrbracket, 0) = 0$. Hence since $\mathfrak{M} \leq \mathfrak{M}'$, $\llbracket E_t \rrbracket_{\mathfrak{M}}(\llbracket \langle P \rangle \rrbracket, 0) = 0$. So $\mathfrak{M} \models \forall p((\langle P \rangle E_t p) \rightarrow p)$. But also since $\mathfrak{M} \models C$, Interpretability and $\text{HasType}(\langle P \rangle, \langle t \rangle)$ are both true in $\mathfrak{M}$, and hence so is $\exists p(\langle P \rangle E_t p)$.

Putting these together we have $\mathfrak{M} \models \exists p(\langle P \rangle E_t p) \wedge \forall p((\langle P \rangle E_t p) \rightarrow p)$, i.e. $\mathfrak{M} \models \text{True}\langle P \rangle$.

□

For the purposes of the next results, we define a family of functions T^n from sentences to sentences, as follows:

$$\begin{aligned} T^0P &:= P \\ T^{m+1}P &:= \text{True}\langle T^mP \rangle \end{aligned}$$

We let T^n be the smallest theory in $\mathcal{L}$ axiomatised by all the sentences T^mP where $C \vdash P$ and $m \leq n$.

Proposition 22. $C_n = T_n$.

Proof. The base case is trivial since $T^0P = P$. Induction step: suppose $C_n = T_n$, and let $C_{n+1} \vdash P$. Then P is a logical consequence of some sentences $Q_1, \dots, Q_m, \text{True}\langle R_1 \rangle, \dots, \text{True}\langle R_k \rangle$ where each Q_i and R_i is a theorem of C_n and hence of T_n by the induction hypothesis. Each Q_i is also a theorem of T_{n+1} since $T_n \subseteq T_{n+1}$. Each R_i is a logical consequence of some axioms $A_{i,1}, \dots, A_{i,k_i}$ of T_n , where each $A_{i,j}$ is of the form T^mP for some $m \leq n$ and C -theorem P . Then $\text{True}\langle A_{i,1} \rangle, \dots, \text{True}\langle A_{i,k_i} \rangle$ are axioms of T_{n+1} . But by Proposition 13, $\text{True}\langle A_{i,1} \rangle \wedge \dots \wedge \text{True}\langle A_{i,k_i} \rangle \rightarrow \text{True}\langle R_i \rangle$ is a theorem of C and hence of T_{n+1} . Hence $\text{True}\langle R_i \rangle$ is also a theorem of T_{n+1} for each i , and so P is a theorem of T_{n+1} too. $\square$

Proposition 23. $\mathfrak{M}_n \models T^mP$ whenever $C \vdash P$ and $m \leq n$.

Proof. $n = 0$: from Corollary 17.

Induction step: suppose for induction that $\mathfrak{M}_n \models T^mP$, whenever $m \leq n$ and $C \vdash P$. We first show that this is also true for any $\mathfrak{M} \in T_{\mathfrak{M}_n}$, i.e. such that $\mathfrak{M} \leq \mathfrak{M}_n$ and $\mathfrak{M} \models C$. When $m = 0$ this is immediate since $\mathfrak{M} \models C$. When $m = k + 1$ for some k , T^mP is $\text{True}\langle T^kP \rangle$, which is true in $\mathfrak{M}$ by Proposition 21 since it's true in $\mathfrak{M}_n$, $\mathfrak{M} \leq \mathfrak{M}_n$, and $\mathfrak{M} \models C$.

To complete the proof we need to show that whenever $m \leq n + 1$ and $C \vdash P$, $\mathfrak{M}_{n+1} \models T^mP$. When $m = 0$, this follows immediately from Corollary 17. Otherwise, $m = k + 1$ for some $k \leq n$, and by what we just showed, T^kP is true in every $\mathfrak{M} \in T_{\mathfrak{M}_n}$. But then by Proposition 20, $\text{True}\langle T^kP \rangle$, i.e. T^mP , is true in $\mathfrak{M}_{T_{\mathfrak{M}_n}}$, i.e. $\mathfrak{M}_{n+1}$. $\square$

From the last two lemmas we have:

Proposition 24. $\mathfrak{M}_n \models C_n$

Since C_n^+ is the result of adding all DE-instances to C_n , and every DE-instance is true in each $\mathfrak{M}_n$ by Proposition 19, we can strengthen this to

Proposition 25. $\mathfrak{M}_n \models C_n^+$.

It follows from the standardness of $\mathfrak{M}_n$ that each C_n^+ is ω -consistent.

[ONE MORE WORTHWHILE FACT: C_n^+ does not prove $T^{n+2}P$ for any P , since $T^{n+2}P$ is not true in $\mathfrak{M}_n$.]

11 Lambda languages 1: with Functionality

12 Lambda languages 2: without Functionality

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